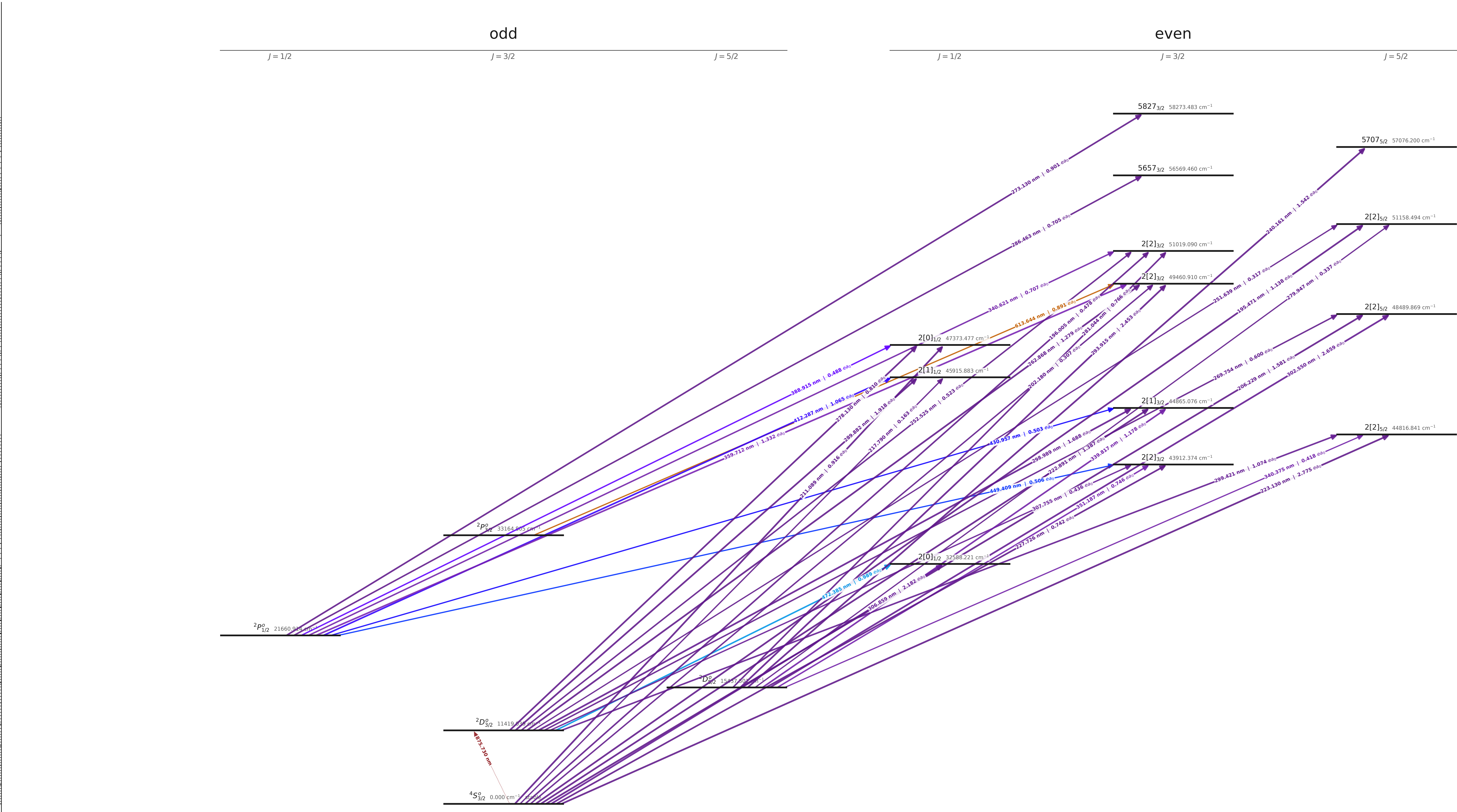


Bismuth

Bi (Bi I). Strongest classified transitions below 2 μm (NIST): 18 levels, 38 lines.

Bi⁺ ionisation limit 58761.65 cm⁻¹ = 7.28552 eV (170.179 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁴ S _{3/2} – 2[2] _{5/2}	195.4710	–	1533.6931	1.138	5.860e+07	–	–	–
⁴ S _{3/2} – 2[2] _{3/2}	206.2286	206.1628	1453.6897	1.581	9.620e+07	–	–	–
⁴ S _{3/2} – 2[0] _{1/2}	211.0886	211.0217	1420.2211	0.916	9.040e+07	–	–	–
⁴ S _{3/2} – 2[1] _{3/2}	222.8905	222.8212	1345.0211	1.387	8.800e+07	–	–	–
⁴ S _{3/2} – 2[2] _{1/2}	223.1304	223.0611	1343.5751	2.775	2.340e+08	–	–	–
⁴ S _{3/2} – 2[2] _{3/2}	227.7262	227.6559	1316.4598	0.742	2.360e+07	–	–	–
² D _{5/2} – 5707 _{5/2}	240.1612	240.0881	1248.2968	1.542	5.800e+07	–	–	–
² D _{3/2} – 2[2] _{1/2}	262.8682	262.7899	1140.4666	1.279	4.560e+07	–	–	–
² P _{1/2} ^o – 5827 _{3/2}	273.1302	273.0494	1097.6172	0.901	2.020e+07	–	–	–
² D _{5/2} – 2[0] _{1/2}	278.1298	278.0477	1077.8869	0.810	3.090e+07	–	–	–
² D _{3/2} – 2[1] _{1/2}	289.8816	289.7967	1034.1894	1.918	1.530e+08	–	–	–
² D _{5/2} – 2[2] _{3/2}	293.9153	293.8294	1019.9961	2.453	1.200e+08	–	–	–
² D _{3/2} – 2[1] _{3/2}	298.9891	298.9019	1002.6870	1.688	5.400e+07	–	–	–
² D _{5/2} – 2[2] _{5/2}	302.5502	302.4621	990.8851	2.659	8.620e+07	–	–	–
⁴ S _{3/2} – 2[0] _{1/2}	306.8593	306.7702	976.9703	2.182	1.670e+08	–	–	–
² P _{1/2} ^o – 2[2] _{3/2}	359.7123	359.6097	833.4229	1.332	1.930e+07	–	–	–

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ_u).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

